Real-Time Anomaly Detection in Financial Transactions

A Performance Analysis Report on Unsupervised Anomaly Detection Models for Credit Card Fraud Detection

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1. Executive Summary

Financial fraud represents one of the most persistent and economically damaging forms of digital crime, and the detection of fraudulent card transactions in near real time remains an open engineering challenge. This project, titled *Real-Time Anomaly Detection in Financial Transactions*, investigates the feasibility of identifying fraudulent credit card transactions without relying on labelled fraud examples during training. The work is motivated by a practical constraint faced by production fraud systems: labels are scarce, delayed, and frequently unreliable, whereas the profile of "normal" customer behaviour is abundantly available. Unsupervised anomaly detection therefore offers an attractive alternative to conventional supervised classification.

Two established unsupervised algorithms were implemented and evaluated on the publicly available Credit Card Fraud Detection dataset: **Isolation Forest**, a tree-ensemble method that isolates anomalies through recursive random partitioning, and **Local Outlier Factor (LOF)**, a density-based method that scores points according to the local density deviation with respect to their neighbours. Both models were trained predominantly on normal transactions in order to learn a representation of legitimate behaviour, and were then evaluated against the full labelled test population.

The Isolation Forest model achieved a precision of **0.2305**, a recall of **0.2947**, and an F1 score of **0.2587**, correctly flagging 135 fraudulent transactions while producing 484 false positives. Critically, the model completed training in approximately **1.33 seconds**. The Local Outlier Factor model, configured with `novelty=True` and fitted on normal transactions, failed entirely under this configuration: it recorded a precision, recall, and F1 score of **0.0000**, detecting none of the 492 fraudulent transactions in the evaluation set while still generating 417 false positives. Its training time was approximately **103.20 seconds**, roughly seventy-seven times slower than the Isolation Forest.

A contamination sweep over the values 0.0017, 0.003, 0.005, and 0.01 confirmed the expected precision–recall trade-off: increasing the assumed contamination rate raised recall from 0.2947 to 0.5122 but degraded precision from 0.2305 to 0.0814, collapsing the F1 score from 0.2587 to 0.1405. The value **0.0017**, which closely mirrors the empirical fraud prevalence in the dataset, produced the highest F1 score and was therefore selected as the operating threshold.

On the combined evidence of detection quality, computational cost, and scalability, the report recommends **Isolation Forest** as the preferred unsupervised detector for highly imbalanced fraud detection workloads, and identifies the Local Outlier Factor as unsuitable for deployment at this data scale in its current configuration.

2. Introduction

2.1 The Problem of Financial Fraud

Card-not-present and card-present fraud continue to impose substantial direct losses on issuers, acquirers, and merchants, alongside indirect costs arising from chargeback handling, customer attrition, and regulatory exposure. A modern payment network processes transactions at a rate of thousands per second, and each authorisation decision must typically be reached within a few tens of milliseconds. Fraud detection is therefore not solely a statistical problem; it is a systems problem constrained simultaneously by accuracy, latency, and throughput.

Rule-based engines dominated the first generation of fraud controls. Such systems encode expert heuristics — unusual merchant categories, improbable geographic velocity, atypical transaction amounts — into deterministic decision logic. While transparent and auditable, rule engines degrade quickly as adversaries adapt. Fraud is an intrinsically non-stationary phenomenon: the moment a rule becomes effective, it also becomes discoverable, and fraudsters restructure their behaviour to evade it. Maintaining a large rule base is consequently expensive and reactive by construction.

2.2 Why Anomaly Detection

Anomaly detection reframes the problem. Rather than attempting to enumerate the ways in which a transaction may be fraudulent, an anomaly detector learns a compact statistical description of *normal* behaviour and flags observations that deviate materially from it. This inversion carries three practical advantages in the fraud domain.

First, it accommodates extreme class imbalance. Fraudulent transactions constitute a vanishingly small fraction of total volume — in the dataset used here, 492 of 284,807 records, or approximately **0.173 %**. Supervised classifiers trained naively on such distributions tend to converge to the degenerate majority-class solution, achieving misleadingly high accuracy while detecting nothing of interest.

Second, anomaly detection does not require exhaustive fraud labels. In operational settings, a transaction is confirmed as fraudulent only after a customer dispute or an investigator's review, a process that may take weeks. Any supervised model is therefore trained on a label set that is both incomplete and stale.

Third, anomaly detection generalises to previously unseen attack patterns. Because the decision boundary is defined by the geometry of normal data rather than by memorised fraud signatures, novel fraud typologies may still register as outliers even though no comparable example appeared during training.

2.3 Rationale for Unsupervised Learning

Given the above, this project deliberately adopts an unsupervised formulation. The labels present in the dataset are used exclusively for *evaluation*, never for model fitting. This design mirrors the realistic deployment scenario in which a detector must be commissioned before a reliable labelled corpus exists, and it permits an honest assessment of how much fraud signal can be recovered from the structure of normal behaviour alone. The two algorithms selected — Isolation Forest and Local Outlier Factor — were chosen because they represent two fundamentally different inductive biases: isolation-based partitioning versus local density estimation. Comparing them therefore yields insight not only into which model performs better, but into *why* one family of assumptions suits high-dimensional, highly imbalanced financial data better than the other.

2.4 Objectives and Scope

The objectives of the project are to characterise the dataset and its imbalance through exploratory analysis; to prepare the feature space appropriately for distance- and partition-based methods; to implement and tune both detectors; to evaluate them using metrics that remain informative under extreme imbalance; and to issue a defensible recommendation regarding model selection and operating threshold. Deep generative models, supervised ensembles, and full streaming deployment lie outside the scope of the present work and are discussed instead as future extensions.

3. Dataset Description

3.1 Structure

The study uses the Credit Card Fraud Detection dataset, a widely referenced benchmark comprising anonymised card transactions. Each record is described by thirty predictive attributes and one binary target attribute.

Attribute	Type	Description
`Time`	Numeric (seconds)	Elapsed seconds between the given transaction and the first transaction in the dataset.
`Amount`	Numeric (currency)	Monetary value of the transaction.
`V1–V28`	Numeric (continuous)	Principal components obtained by PCA transformation of the original, confidential transaction attributes.
`Class`	Binary	Ground-truth label: `0` denotes a normal transaction, `1` denotes a fraudulent transaction.

The evaluation population contains **284,315 normal transactions** and **492 fraudulent transactions**, for a total of **284,807 records**.

3.2 The PCA-Transformed Feature Space

Features `V1` through `V28` are not raw observables. The dataset publishers applied Principal Component Analysis to the original attribute set — which plausibly included merchant identifiers, geographic indicators, device fingerprints, and cardholder attributes — in order to satisfy confidentiality obligations. Three consequences of this transformation are material to the present analysis.

The components are, by construction, **mutually uncorrelated** and ordered by decreasing explained variance. This is advantageous for distance-based and partition-based detectors, as it removes the redundancy that would otherwise cause correlated dimensions to be implicitly over-weighted. It also means that no further decorrelation step is required during preprocessing.

The components are **already centred and comparably scaled**, having been produced from a standardised input space. `Time` and `Amount`, by contrast, were released in their original units and therefore occupy numeric ranges that are orders of magnitude larger than those of the principal components. This scale mismatch is the principal preprocessing concern addressed in Section 5.

Finally, the components are **semantically opaque**. No component can be mapped back to an interpretable business quantity, which precludes domain-driven feature engineering and limits the interpretability of any explanation offered to a fraud analyst. Model outputs must therefore be justified statistically rather than causally.

3.3 Severe Class Imbalance

The defining statistical property of this dataset is its imbalance. With 492 positives against 284,315 negatives, the positive class prevalence is approximately **0.1727 %**, a ratio of roughly one fraudulent transaction for every 578 legitimate ones.

This imbalance has direct methodological implications. A trivial classifier that labels every transaction as normal attains an accuracy of **99.83 %** while providing zero operational value. Accuracy is therefore rejected as an evaluation metric throughout this report. Furthermore, because only 492 positives exist in total, every additional true positive shifts recall by approximately 0.2 percentage points, and metric estimates carry non-trivial variance. Conclusions are consequently drawn from the *relative* ordering of models and from the *shape* of the precision–recall trade-off rather than from small absolute differences in any single statistic.

Imbalance also governs the interpretation of false positives. Because the negative class is so large, even an exceptionally low false-positive *rate* translates into a substantial false-positive *count*. A false-positive rate of 0.17 % applied to 284,315 normal transactions yields approximately 484 flagged legitimate transactions — precisely the figure observed for the Isolation Forest. This arithmetic, rather than any deficiency in the algorithm, is the fundamental reason why precision remains low in fraud detection systems.

4. Data Preprocessing

4.1 Exploratory Data Analysis

Exploratory analysis preceded all modelling and served three purposes: verifying data integrity, quantifying the imbalance, and characterising the distributions of the two unscaled features.

The **class distribution** was examined first and confirmed the 492-to-284,315 split described above. Visualised on a linear axis, the fraudulent class is effectively invisible, and a logarithmic count axis was required to render both classes legibly. This visualisation established at the outset that resampling-free unsupervised methods and imbalance-robust metrics would be necessary.

The **Amount distribution** was found to be strongly right-skewed, with a dense concentration of low-value transactions and a long tail extending to comparatively large values. The distribution is heavy-tailed rather than Gaussian, and a small number of extreme legitimate purchases occupy the upper tail. This finding directly motivated the choice of scaler discussed below.

The **Time distribution** exhibited a clear cyclical structure consistent with diurnal human activity: transaction density rises during waking hours and falls markedly during overnight periods, producing a recognisable multi-modal profile across the observation window. Because `Time` is measured as an absolute offset from the first record rather than as a time-of-day value, this cyclicality is present but not directly encoded as a periodic feature — a limitation revisited in Section 10.

No missing values were identified in any attribute, and no duplicate-driven data integrity issues required remediation prior to modelling.

4.2 Robust Scaling of `Time` and `Amount`

Both algorithms under study are sensitive to feature scale, though for different reasons. LOF computes explicit distances between neighbouring points, so any feature with a disproportionately large numeric range will dominate the distance metric and effectively suppress the contribution of all other dimensions. Isolation Forest is less directly affected, since axis-aligned random splits are scale-invariant per feature, but the *distribution* of split values and hence the expected path length is still shaped by each feature's spread. Leaving `Time` and `Amount` unscaled would therefore distort both models.

`RobustScaler` was selected in preference to standardisation or min–max normalisation. It centres each feature on its median and scales by the interquartile range:

$$x_{scaled} = (x - median(x))/(Q_3(x) - Q_1(x))$$

This choice is deliberate and follows directly from the exploratory findings. `StandardScaler` relies on the mean and standard deviation, both of which are strongly influenced by extreme values; applied to the heavy-tailed `Amount` distribution, it would compress the bulk of the data into a narrow band around zero and thereby reduce the resolution available for distinguishing ordinary from unusual amounts. `MinMaxScaler` is even more fragile, as it maps the single largest observation to the boundary of the output range and rescales all remaining values relative to that outlier. Because the median and interquartile range are order statistics, `RobustScaler` is materially unaffected by the tail. This property is essential in an anomaly detection pipeline, where the extreme observations are not noise to be suppressed but the very signal the detector must remain sensitive to.

4.3 Rationale for Scaling Only Two Features

Only `Time` and `Amount` were scaled; `V1` through `V28` were left untouched. This is a consequence of the dataset's construction rather than an arbitrary decision. The principal components were derived from an already standardised feature space and are therefore approximately zero-centred with comparable, moderate variances. Applying a second scaling transformation to them would provide no benefit and would carry two risks: it would rescale each component independently and thereby destroy the variance ordering that PCA establishes, and it would inflate the relative influence of low-variance components that carry little information. Confining the transformation to the two features actually expressed in original units brings the entire feature matrix onto a consistent footing with the minimum necessary intervention.

4.4 Training Configuration

Consistent with the unsupervised framing, both models were fitted primarily on **normal transactions**, so that each learned a description of legitimate behaviour rather than a discriminative boundary between two labelled classes. Labels were withheld during fitting and used only to compute evaluation metrics on the full population. For LOF this configuration required the `novelty=True` setting, which converts the estimator from a transductive outlier-detection tool into an inductive novelty detector capable of scoring previously unseen points via `predict`. The consequences of this configuration are examined in Section 6.

5. Methodology

5.1 Isolation Forest

Isolation Forest departs from the conventional assumption that anomalies must be identified by first modelling normality in full. Instead it exploits a simpler observation: anomalies are *few and different*, and points that are few and different are easy to separate from the rest of the data by random partitioning.

The algorithm constructs an ensemble of *isolation trees*. Each tree is grown on a subsample of the data by repeatedly selecting a feature at random and a split value at random within that feature's observed range, recursively partitioning until every point is isolated in its own leaf or a height limit is reached. The number of edges traversed from the root to the leaf containing a given point is its **path length**. Because anomalous points lie in sparse regions of the feature space, a random split is comparatively likely to separate them early, producing short path lengths. Points embedded in dense regions require many successive splits before isolation and therefore yield long path lengths.

Averaging the path length $h(x)$ over all trees and normalising against the expected path length $c(n)$ of an unsuccessful search in a binary search tree of n points yields the anomaly score:

$$s(x, n) = 2^{-(E[h(x)])/(c(n))}$$

Scores approaching 1 indicate strong anomalies, scores substantially below 0.5 indicate normal points. The `contamination` parameter then converts this continuous score into a binary decision by fixing the proportion of the data to be labelled anomalous, which is equivalent to selecting a percentile cut-off on the score distribution.

Two properties make this approach well suited to the present problem. Its complexity is approximately linear in the number of samples and it operates on subsamples, giving very low training cost. And because it never computes pairwise distances or densities, it is largely insensitive to the concentration-of-distance effects that impair neighbourhood methods in higher dimensions.

The hyperparameter `n_estimators` was tuned over the candidate values 50, 100, and 200, with **50** selected. That the smallest ensemble sufficed indicates that the path-length estimates stabilise rapidly on this dataset, and that additional trees contributed variance reduction without improving the ranking of anomalies. The `contamination` parameter was tuned separately and is treated in detail in Section 8.

5.2 Local Outlier Factor

Local Outlier Factor takes a density-based view. Rather than assigning a single global notion of outlyingness, it evaluates each point against the density of its own neighbourhood, so that a point in a naturally sparse region is not penalised merely for being sparse.

For a point p and a chosen neighbourhood size k, the *reachability distance* of p with respect to a neighbour o is defined as the maximum of the k-distance of o and the actual distance between the two points. The *local reachability density* of p is the reciprocal of the mean reachability distance over its k nearest neighbours. The Local Outlier Factor is then the ratio of the average local reachability density of

p's neighbours to that of p itself:

$$LOF_k(p) = (1)/(|N_k(p)|)\Sigma_{o \in N_k(p)} (lrd_k(o))/(lrd_k(p))$$

A value near 1 indicates that p is as densely surrounded as its neighbours and is therefore normal. Values appreciably greater than 1 indicate that p resides in a region of lower density than its neighbours and is therefore locally anomalous.

In the standard configuration LOF is transductive: it scores only the points on which it was fitted and exposes no mechanism for scoring new observations. Since this project requires a detector that can be trained on normal data and then applied to unseen transactions, LOF was instantiated with `novelty=True` and fitted on the normal population, after which `predict` was used to classify the evaluation set. The computational cost of the method is dominated by nearest-neighbour search, which scales super-linearly with sample count and degrades further as dimensionality increases and spatial indexing structures lose effectiveness.

5.3 Evaluation Metrics

Given the imbalance established in Section 3.3, accuracy is excluded from consideration. The following metrics, all computed with fraud as the positive class, form the basis of the evaluation.

Precision measures the reliability of a fraud alert, *TP / (TP + FP)*. It is the metric most directly aligned with operational cost, since every false positive consumes investigator time and risks blocking a legitimate customer transaction.

Recall measures coverage, *TP / (TP + FN)*. It corresponds to the proportion of fraudulent value the system actually intercepts, and each false negative represents a realised financial loss.

F1 score is the harmonic mean of precision and recall, *2PR/(P+R)*. Because the harmonic mean is dominated by the smaller of its arguments, F1 cannot be inflated by optimising one metric at the expense of the other and therefore serves as the primary model-selection criterion in this study.

The **confusion matrix** is reported in full for both models, as it exposes the absolute counts that ratio metrics obscure — most importantly the raw volume of false positives, which determines whether a configuration is operationally viable at all.

The **precision–recall curve** was plotted for both detectors. Precision–recall analysis is strongly preferred to ROC analysis under severe imbalance, because the false-positive rate that forms the ROC abscissa is normalised by an enormous negative class and consequently remains small even when the absolute number of false alarms is unacceptable, producing optimistic curves.

Training time is recorded as a first-class metric rather than an implementation footnote, since the stated objective is real-time detection and any candidate model must be retrainable at a cadence matching the drift rate of fraud behaviour.

6. Experimental Results

6.1 Isolation Forest

Under its selected configuration — `n_estimators = 50` and `contamination = 0.0017` — the Isolation Forest produced the following results.

Metric	Value
Precision	0.2305
Recall	0.2947
F1 Score	0.2587
Training Time	≈ 1.33 seconds

The corresponding confusion matrix is:

$$\begin{bmatrix} 283831 & 484 \\ 357 & 135 \end{bmatrix}$$

	Predicted Normal	Predicted Fraud
Actual Normal	283,831 (TN)	484 (FP)
Actual Fraud	357 (FN)	135 (TP)

The model identified **135 fraudulent transactions** while raising **484 false alarms** and missing **357 frauds**. Several observations follow from these counts.

The false-positive rate is *484 / 284,315 ≈ 0.170 %*, which corresponds almost exactly to the specified contamination of 0.0017. This is expected behaviour: contamination acts as a percentile cut-off on the anomaly score, so the volume of flagged transactions is fixed by the parameter and the model's task is to ensure that the flagged set is enriched with genuine fraud.

That enrichment is substantial. Random selection of 619 transactions from the population would be expected to contain approximately one fraud, given a base rate of 0.173 %. The model's flagged set contains 135. The detector therefore concentrates fraud in its alert queue by more than two orders of magnitude relative to chance — a result that a precision figure of 0.2305, read in isolation, considerably understates.

Overall accuracy on the evaluation population is *(283,831 + 135)/284,807 ≈ 99.71 %*, which is *lower* than the 99.83 % attainable by predicting the majority class everywhere. This inversion is a useful illustration of why accuracy was excluded from the metric set: the model trades a small amount of accuracy for the detection of 135 frauds that the trivial classifier would miss entirely.

Training completed in approximately **1.33 seconds** on nearly 285,000 records. This confirms the theoretical near-linear scaling of the algorithm and establishes that frequent retraining — hourly or even more often — is computationally trivial.

The precision–recall curve for the Isolation Forest is monotonically decreasing and lies clearly above the no-skill baseline of 0.0017 across the low-recall region, confirming that the underlying anomaly score carries genuine discriminative information and that the observed operating point reflects a real ranking capability rather than an artefact of threshold placement.

6.2 Local Outlier Factor

The Local Outlier Factor, configured with `novelty=True` and fitted on normal transactions, produced the following results.

Metric	Value
Precision	0.0000
Recall	0.0000
F1 Score	0.0000
Training Time	≈ 103.20 seconds

The corresponding confusion matrix is:

$$\begin{bmatrix} 283898 & 417 \\ 492 & 0 \end{bmatrix}$$

	Predicted Normal	Predicted Fraud
Actual Normal	283,898 (TN)	417 (FP)
Actual Fraud	492 (FN)	0 (TP)

The result constitutes a complete detection failure. All **492 fraudulent transactions** were classified as normal, yielding zero true positives and hence zero precision, zero recall, and zero F1. The model nevertheless flagged **417 legitimate transactions** as anomalous, so it did not simply predict a constant class; it produced an alert budget of comparable size to the Isolation Forest's and expended the entirety of that budget on legitimate transactions.

This is the more damaging interpretation of the two available. A degenerate model that flags nothing is at least harmless. A model that flags 417 transactions and achieves a precision of exactly zero has an alert queue that is not merely uninformative but actively *worse* than random selection, since random selection of 417 transactions would be expected to include roughly one genuine fraud.

Several factors plausibly contribute to this outcome. LOF's novelty mode constructs its density model exclusively from the normal training manifold, and the decision boundary is drawn at the periphery of that manifold according to the contamination setting. If fraudulent transactions in this thirty-dimensional PCA space do not occupy a region of markedly lower local density than the sparse outer regions of legitimate behaviour, the boundary will preferentially capture unusual-but-legitimate transactions — for example, the extreme values in the heavy right tail of `Amount` identified during exploratory analysis. In addition, LOF's behaviour is highly sensitive to the neighbourhood size k, which governs the spatial scale at which density is assessed; a value poorly matched to the local structure of a very large, dense normal population will misplace the boundary. Finally, in thirty dimensions the contrast between nearest and farthest neighbour distances contracts, so the density ratios on which LOF depends lose the dynamic range required to separate classes. The result should therefore be read as a failure of *this configuration* of LOF on *this feature space*, and it demonstrates that a sound theoretical foundation does not guarantee empirical performance without geometry-aware tuning.

Training required approximately **103.20 seconds**, some **seventy-seven times** the Isolation Forest's cost, reflecting the expense of nearest-neighbour computation over a large, high-dimensional sample.

The precision–recall curve for LOF is correspondingly uninformative, remaining flat at or near the no-skill baseline and offering no threshold at which a useful precision–recall combination is attainable. This is an important diagnostic: it establishes that the failure is not a matter of an unfortunate threshold choice but a deficiency in the underlying score ranking itself. No re-thresholding of this model would recover usable performance.

6.3 Execution Time and Visual Analysis

The execution-time comparison chart renders the disparity starkly, with the LOF bar exceeding the Isolation Forest bar by nearly two orders of magnitude. In a system targeting real-time operation, this gap is decisive.

The PCA scatter plot, which projects the feature space onto its two leading components with fraudulent transactions overlaid, clarifies the difficulty both models face. Fraudulent points are not confined to a single compact, well-separated cluster; a portion of them lie in or adjacent to the dense body of legitimate transactions. Fraud in this dataset is therefore only *partially* separable in the available feature space, which places an upper bound on the performance attainable by any purely geometric unsupervised detector and contextualises the Isolation Forest's modest but genuine 0.2587 F1 score.

The confusion-matrix heatmaps for both models are dominated visually by the true-negative cell, a direct consequence of the imbalance, and the performance comparison bar chart summarises the precision, recall, and F1 results side by side, with all three LOF bars at zero.

7. Comparative Analysis

7.1 Summary Comparison

Criterion	Isolation Forest	Local Outlier Factor
Precision	0.2305	0.0000
Recall	0.2947	0.0000
F1 Score	0.2587	0.0000
True Positives	135	0
False Positives	484	417
False Negatives	357	492
True Negatives	283,831	283,898
Training Time	≈ 1.33 s	≈ 103.20 s
Algorithmic Basis	Random partitioning / path length	Local density ratio
Approximate Complexity	Near-linear in n (subsampled)	Super-linear in n (neighbour search)
Deployment Suitability	Suitable	Not suitable as configured

7.2 Precision

Neither model achieves high precision, but the distinction between them is categorical rather than incremental. At 0.2305, the Isolation Forest returns approximately one genuine fraud for every four alerts. Measured against a base rate of 0.173 %, this represents an enrichment factor above 130, and it is a workable figure for a system in which alerts are triaged by an investigation team or used to trigger step-up authentication rather than an outright block. LOF's precision of 0.0000 admits no such interpretation: its alert queue contains no actionable items whatsoever, and any downstream process consuming it would waste its entire capacity.

7.3 Recall

The Isolation Forest recovered 0.2947 of fraudulent transactions, intercepting 135 of 492. This is far from complete coverage and leaves 357 frauds undetected, so the model would necessarily operate as one layer within a defence-in-depth architecture rather than as a sole control. Nevertheless it establishes a meaningful unsupervised baseline achieved without a single fraud label at training time. LOF's recall of 0.0000 provides no coverage at all and would leave the institution's exposure entirely unmitigated.

7.4 F1 Score

Because F1 is a harmonic mean, it is bounded above by twice the smaller of its arguments and cannot be gamed by extremising one component. The Isolation Forest's 0.2587 sits close to the geometric centre of its precision and recall, indicating a balanced operating point rather than a skewed one. LOF's F1 of 0.0000 is a mathematical necessity once precision and recall both vanish and conveys no additional information beyond confirming total failure.

7.5 Training Time and Computational Complexity

The 1.33-second versus 103.20-second contrast is the single most operationally consequential difference between the two models, and it reflects a structural rather than an incidental gap.

Isolation Forest trains each tree on a small subsample and performs only random axis-aligned splits, giving a cost that is approximately *O(t · psi log psi)* for *t* trees and subsample size *psi* — effectively independent of total dataset size once *psi* is fixed. Scoring a single new transaction requires traversing 50 shallow trees, an operation measurable in microseconds and comfortably within a real-time authorisation budget. The model is also embarrassingly parallel across trees.

LOF must construct a neighbourhood structure over the entire training population and, in novelty mode, retain that structure so that new points can be positioned within it. Nearest-neighbour queries degrade towards linear scans as dimensionality rises and spatial indices lose their pruning ability, so cost grows super-linearly in *n* and unfavourably in *d*. Memory consumption is correspondingly higher, since the training data must be retained rather than compressed into a model. At thirty dimensions and 284,000 records this dataset already sits in the regime where these costs bite; at production volumes of tens of millions of transactions per day the approach would be infeasible.

The practical significance extends beyond a single training run. Fraud patterns drift continuously, so a deployed detector must be retrained frequently. A 1.33-second training cost permits retraining on a rolling window many times per hour at negligible expense. A 103-second cost, multiplied across model variants, validation folds, and hyperparameter searches, turns routine maintenance into a scheduled batch job and materially slows the iteration cycle.

7.6 False Positives

The two models raised comparable volumes of false alarms — 484 for Isolation Forest and 417 for LOF — but the value delivered alongside them differs absolutely. The Isolation Forest's 484 false positives accompany 135 true positives, so an investigation team working the queue encounters an actionable case roughly every fourth item. LOF's 417 false positives accompany none. LOF's marginally lower false-positive count is therefore not a point in its favour; it is a smaller volume of entirely wasted work.

It is worth emphasising that a false-positive rate of 0.170 % would be considered excellent in most classification settings. Its translation into 484 absolute alerts is purely a consequence of the size of the negative class, and this arithmetic — not algorithmic weakness — is the dominant constraint on precision in fraud detection. Operationally, this is why alert volume is budgeted against investigator capacity and why the contamination parameter functions in practice as a capacity control.

7.7 False Negatives

The Isolation Forest missed 357 frauds, LOF missed all 492. Since false negatives correspond directly to realised losses, the Isolation Forest prevented losses on 135 transactions that LOF did not. The asymmetry between error types is important here: a false positive costs a review and some customer friction, whereas a false negative costs the full fraudulent transaction amount plus chargeback handling. Under most plausible cost matrices for this domain, the false-negative cost dominates by an order of magnitude or more, which strengthens the case for the Isolation Forest and argues for revisiting the threshold in Section 8 whenever institutional cost estimates justify trading precision for recall.

7.8 Practical Deployment Suitability

Assessed against the requirements of a real-time fraud pipeline, the Isolation Forest is viable. It trains in seconds, scores in microseconds, parallelises naturally, retains a compact model artefact independent of training-set size, exposes a single interpretable operating parameter in `contamination`, and delivers a fraud-enriched alert queue without requiring labels. Its 0.2947 recall means it must be deployed as one signal among several — realistically combined with a rule engine, velocity checks, and, once labels accumulate, a supervised model — but it functions correctly in that role.

LOF, in the configuration evaluated here, is not deployable. Its detection performance is nil, its training cost is two orders of magnitude higher, its memory footprint scales with the data, and its precision–recall curve indicates that no threshold adjustment would rescue it. Recovering value from LOF would require dimensionality reduction to a lower-dimensional subspace, systematic tuning of the neighbourhood parameter, and possibly restriction to a sampled or clustered subset of the normal population — work that falls outside the present scope.

7.9 Why Isolation Forest Outperformed LOF

The performance gap traces to the interaction between each algorithm's inductive bias and the specific geometry of this dataset.

The decisive factor is **dimensionality**. LOF depends on density ratios computed from neighbour distances, and in thirty dimensions the distances from a query point to its nearest and farthest neighbours converge, compressing the dynamic range of the density estimate until fraudulent and legitimate points become indistinguishable by that measure. Isolation Forest requires no distance computation. It needs only that *some* feature, in *some* tree, tends to separate an anomaly early — a far weaker requirement that random feature selection satisfies robustly as dimensionality grows.

Locality assumptions compound the problem. LOF is designed to detect points that are anomalous relative to their immediate neighbourhood, which is powerful when normal data comprises multiple clusters of differing density. In this dataset the normal population forms one very large, dense mass. Within such a structure, the points with the lowest local density are the legitimate outliers in the tails — including the high-value transactions visible in the `Amount` distribution — and it is precisely these that LOF's boundary captured, as its 417 false positives with zero true positives confirms.

Class overlap limits both models, as the PCA scatter plot shows, but it penalises the density-based method more severely. Where fraudulent points sit inside the dense legitimate mass, their local density is comparable to their neighbours' and their LOF score approaches 1, the canonical "normal" value. Isolation Forest can still isolate such points early if any single feature places them at an unusual value, giving it a route to detection that LOF lacks.

Global versus local scoring is the final consideration. Isolation Forest produces a global anomaly ranking calibrated against the whole dataset, which aligns naturally with a contamination-based threshold set to the known fraud prevalence. LOF's scores are locally normalised by construction, so a single global cut-off applied to them lacks a consistent interpretation across regions of differing density.

Taken together, these factors explain why the simpler and computationally cheaper method is also the substantially more effective one here. The outcome is a concrete instance of a general principle: in high-dimensional anomaly detection, methods that avoid explicit distance and density estimation tend to be more robust than those that depend on them.

8. Threshold Recommendation

8.1 Contamination Tuning Results

The `contamination` parameter of the Isolation Forest does not alter the learned model; it selects the percentile of the anomaly score distribution above which a transaction is declared fraudulent. It is therefore the primary operating control, and its calibration determines the balance between fraud captured and legitimate transactions disturbed. Four values were evaluated.

Contamination	Precision	Recall	F1 Score
0.0017	**0.2305**	**0.2947**	**0.2587**
0.003	0.1613	0.3333	0.2174
0.005	0.1179	0.3862	0.1806
0.010	0.0814	0.5122	0.1405

8.2 Interpretation of the Trade-Off

The sweep exhibits the canonical precision–recall trade-off with unusual clarity. Raising contamination from 0.0017 to 0.010 — a roughly six-fold expansion of the alert budget — increased recall from 0.2947 to 0.5122, an improvement of about 74 %. Over the same interval precision fell from 0.2305 to 0.0814, a deterioration of about 65 %. Because precision declined proportionally faster than recall improved, the F1 score fell monotonically from 0.2587 to 0.1405, a loss of some 46 %.

The mechanism is straightforward. The anomaly score ranks transactions by outlyingness, and fraud is concentrated towards the top of that ranking. As the threshold is relaxed to admit progressively lower-scoring transactions, each successive tranche is less enriched with fraud than the one before it, because the strongest fraud signals were already captured. Meanwhile the number of legitimate transactions admitted grows in proportion to the enormous negative class. Each increment of contamination therefore purchases a diminishing quantity of additional true positives at a steadily rising cost in false positives.

The arithmetic makes the effect concrete. At a contamination of 0.010, approximately 2,848 transactions are flagged. Of these, roughly 252 would be genuine frauds at the observed precision of 0.0814, leaving about 2,596 legitimate transactions requiring review — more than five times the false-positive volume at the selected threshold, in exchange for approximately 117 additional detections.

8.3 Why Higher Contamination Increases False Positives

Two structural facts drive this behaviour. First, contamination is defined as a proportion of the *entire* dataset, and that dataset is 99.83 % legitimate; consequently the overwhelming majority of any newly admitted tranche is legitimate by construction, irrespective of how well the model ranks. Second, the pool of available true positives is capped at 492, whereas the pool of available false positives numbers

284,315. Once the model has surfaced the fraudulent transactions that are genuinely separable in the feature space — and Section 6.3 indicates that many are not cleanly separable — further relaxation of the threshold draws almost exclusively from the negative class. Precision therefore decays towards the base rate of 0.0017 as contamination grows.

8.4 Recommendation

Contamination = 0.0017 is recommended as the default operating threshold.

The justification rests on three grounds. It attains the highest F1 score of the values tested, 0.2587, and F1 is the appropriate selection criterion because its harmonic form prevents either metric from being sacrificed for the other. It corresponds closely to the empirical fraud prevalence of approximately 0.173 %, meaning the model's alert budget is matched to the true rate of the phenomenon rather than to an arbitrary assumption — a principled calibration and not a coincidence. And it produces a manageable alert volume of 619 flagged transactions containing 135 genuine frauds, a queue that a modest investigation function can process while encountering an actionable case roughly one time in four.

This recommendation should nevertheless be qualified as *cost-neutral*. F1 implicitly weights precision and recall equally, whereas the true cost of a false negative in payments — the transaction value plus chargeback and remediation overhead — typically exceeds the cost of a false positive, which is an investigator's time plus some customer friction. Where an institution can quantify these costs, the threshold should be re-selected by minimising expected cost rather than by maximising F1, and such an analysis would generally shift the optimum towards higher contamination and higher recall. A value of 0.003, which raises recall to 0.3333 while retaining a precision of 0.1613, represents a reasonable intermediate posture for an organisation with additional review capacity and a high assessed cost of undetected fraud.

In practice the threshold should be treated as a tunable operational control rather than a fixed hyperparameter. It should be reviewed periodically against realised fraud losses and current investigator capacity, and it may reasonably be varied by segment — applying a more aggressive threshold to high-value or high-risk transaction categories where the expected loss per missed case is greatest, and a more conservative one to low-value flows where customer friction dominates.

9. Conclusion

This project set out to determine how effectively fraudulent credit card transactions can be identified by unsupervised anomaly detection, using no fraud labels during model fitting. Two algorithms embodying distinct inductive biases were implemented, tuned, and evaluated on the Credit Card Fraud Detection dataset, and the results permit clear conclusions.

Isolation Forest achieved the best overall performance. With `n_estimators = 50` and `contamination = 0.0017`, it recorded a precision of 0.2305, a recall of 0.2947, and an F1 score of 0.2587, correctly identifying 135 of 492 fraudulent transactions while flagging 484 legitimate ones. Although the absolute precision is modest, it represents an enrichment of fraud in the alert queue exceeding two orders of magnitude over the 0.173 % base rate, and it was obtained without any supervision. Training completed in approximately 1.33 seconds, and per-transaction scoring reduces to traversal of 50 shallow trees, making the model comfortably compatible with real-time authorisation latency budgets and with frequent retraining as fraud behaviour drifts.

Local Outlier Factor was significantly slower and failed to detect fraud effectively. Configured with `novelty=True` and fitted on normal transactions, it returned a precision, recall, and F1 score of 0.0000, detecting none of the 492 frauds while still flagging 417 legitimate transactions — an alert queue of comparable size to the Isolation Forest's containing nothing of value. Its training time of approximately 103.20 seconds was some seventy-seven times greater. The flat precision–recall curve confirms that this failure resides in the score ranking itself and could not be remedied by threshold adjustment. The result reflects the well-documented degradation of density-based methods in high-dimensional spaces, compounded by the single dense, non-clustered structure of the normal transaction population, within which the lowest-density points are legitimate tail transactions rather than frauds.

Isolation Forest is therefore recommended for highly imbalanced fraud detection datasets. Its advantage is structural rather than incidental: by avoiding explicit distance and density estimation it sidesteps the geometric pathologies that impair neighbourhood methods, while its near-linear complexity and compact model artefact satisfy the operational demands of a production payment pipeline. The contamination sweep further establishes 0.0017 as the F1-optimal operating point, closely matching the empirical fraud prevalence, with the caveat that a cost-weighted analysis incorporating institution-specific loss estimates would likely justify a somewhat higher setting.

Two broader observations merit record. First, algorithmic sophistication does not predict empirical performance; the simpler and cheaper method comfortably outperformed the more theoretically elaborate one on this data. Second, the ceiling on achievable performance is set substantially by the feature space itself — the PCA projection shows fraudulent and legitimate transactions to be only partially separable — which indicates that the most productive direction for improvement lies in richer behavioural representation rather than in further tuning of the detectors evaluated here.

10. Future Work

The results establish a functional unsupervised baseline and simultaneously identify the constraints that limit it. The following extensions are ordered by expected return relative to implementation effort.

10.1 Feature Engineering and Behavioural Context

The clearest limitation of the present work is representational. The thirty available features describe each transaction in isolation, whereas fraud is often recognisable only in the context of a cardholder's transaction history. Because `Time` is supplied as an absolute offset, its pronounced diurnal cyclicality — evident in the exploratory analysis — is not directly available to the models. Deriving a cyclic time-of-day encoding, for example via sine and cosine transforms, together with a day-of-week indicator, would expose this structure at negligible cost. More substantially, entity-level aggregate features such as rolling transaction counts and amount statistics over recent windows, deviation of the current amount from the cardholder's historical mean, and time elapsed since the previous transaction would convert isolated observations into behavioural trajectories. Given that the PCA scatter plot attributes much of the residual error to class overlap in the current space, enriched features are likely to yield a larger improvement than any change of algorithm.

10.2 Autoencoder-Based Detection

An autoencoder trained exclusively on normal transactions offers a natural extension of the reconstruction-error paradigm to non-linear structure. Compressing the input through a bottleneck forces the network to learn a compact manifold of legitimate behaviour; transactions lying off that manifold reconstruct poorly, and the reconstruction error serves as an anomaly score with a threshold set from the error distribution of held-out normal data. Unlike LOF, this approach requires no distance computation and is therefore not subject to the same dimensional degradation, while unlike Isolation Forest it can capture non-linear interactions among the principal components. Variational and denoising variants merit comparison, the latter being particularly appropriate given the noise characteristics of transactional data.

10.3 Deep and Hybrid Architectures

Beyond the standard autoencoder, several architectures are well matched to the problem. Deep Support Vector Data Description learns a neural embedding in which normal data is enclosed within a minimum-volume hypersphere, providing an end-to-end objective aligned directly with one-class detection. Adversarially trained generative models can be used to score how well a transaction is explained by a learned model of normality. Where labels do accumulate over time, semi-supervised and hybrid designs become attractive: an unsupervised detector can supply a high-recall candidate set which a supervised classifier — gradient-boosted trees, for instance — then re-ranks for precision. Such an architecture leverages available labels without inheriting the cold-start fragility of a purely supervised system.

10.4 Streaming Anomaly Detection

The project title anticipates real-time operation, but the present evaluation is conducted in batch. Genuine streaming detection requires algorithms that update incrementally as data arrives. Half-Space Trees provide an online analogue of the isolation principle with constant-time updates and bounded memory, and Robust Random Cut Forests extend isolation-based detection to streams with explicit support for concept drift. A production design would combine such an online scorer with an explicit drift-detection mechanism and an automated retraining trigger, so that the model adapts as adversarial behaviour changes rather than decaying between scheduled refreshes.

10.5 Real-Time Deployment Architecture

Translating the model into an operational control requires infrastructure beyond the estimator. A realistic architecture would ingest authorisation events through a durable log such as Kafka, compute behavioural aggregates in a low-latency feature store so that training and serving read identical feature definitions, and expose scoring behind a service with a strict latency budget of a few tens of milliseconds. Model artefacts and their contamination settings should be versioned, with shadow deployment and champion–challenger evaluation used to validate candidates against live traffic before promotion. Monitoring must cover not only technical availability but score distribution drift, alert volume against investigator capacity, and realised precision as investigation outcomes return. A feedback channel capturing those outcomes closes the loop and supplies the labels needed for the supervised layer described in Section 10.4.

10.6 Improved Evaluation and Cost-Sensitive Optimisation

Finally, the evaluation itself can be strengthened. Threshold selection should be driven by expected monetary cost rather than F1 once per-error costs are estimated, since the asymmetry between a missed fraud and a false alarm is substantial in this domain. Because only 492 positives are available, metric estimates carry appreciable variance, and reporting confidence intervals via bootstrap resampling would properly characterise that uncertainty. Evaluation should also respect the temporal ordering of transactions, training on earlier periods and testing on later ones, so that reported performance reflects the model's ability to generalise forward in time — the only setting in which it will actually be used. Complementing these, a systematic tuning study of LOF over reduced-dimensionality subspaces and neighbourhood sizes would determine whether the failure documented here is intrinsic to the method on this data or an artefact of the specific configuration evaluated.

References

1. Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." *Proceedings of the Eighth IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.
2. Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation-Based Anomaly Detection." *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 1, 2012.
3. Breunig, M. M., Kriegel, H.-P., Ng, R. T., and Sander, J. "LOF: Identifying Density-Based Local Outliers." *Proceedings of the ACM SIGMOD International Conference on Management of Data*, 2000, pp. 93–104.
4. Chandola, V., Banerjee, A., and Kumar, V. "Anomaly Detection: A Survey." *ACM Computing Surveys*, vol. 41, no. 3, 2009.
5. Aggarwal, C. C. *Outlier Analysis*. 2nd ed., Springer, 2017.
6. Saito, T., and Rehmsmeier, M. "The Precision–Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets." *PLOS ONE*, vol. 10, no. 3, 2015.
7. Dal Pozzolo, A., Caelen, O., Johnson, R. A., and Bontempi, G. "Calibrating Probability with Undersampling for Unbalanced Classification." *IEEE Symposium Series on Computational Intelligence*, 2015.
8. Pedregosa, F. et al. "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, vol. 12, 2011, pp. 2825–2830.
9. Aggarwal, C. C., and Sathe, S. *Outlier Ensembles: An Introduction*. Springer, 2017.
10. Ruff, L. et al. "A Unifying Review of Deep and Shallow Anomaly Detection." *Proceedings of the IEEE*, vol. 109, no. 5, 2021.

Appendix A — Consolidated Results

Isolation Forest — Final Configuration

Parameter	Value
`n_estimators`	50 (selected from 50, 100, 200)
`contamination`	0.0017 (selected from 0.0017, 0.003, 0.005, 0.01)
Training data	Normal transactions
Precision	0.2305
Recall	0.2947
F1 Score	0.2587
Confusion Matrix	[[283831, 484], [357, 135]]
Training Time	≈ 1.33 s

Local Outlier Factor — Final Configuration

Parameter	Value
`novelty`	True
Training data	Normal transactions
Precision	0.0000
Recall	0.0000
F1 Score	0.0000
Confusion Matrix	[[283898, 417], [492, 0]]
Training Time	≈ 103.20 s

Appendix B — Visualisations Produced

Exploratory analysis: class distribution, `Amount` distribution, `Time` distribution. Model evaluation: Isolation Forest confusion-matrix heatmap, LOF confusion-matrix heatmap, PCA scatter plot, Isolation Forest precision–recall curve, LOF precision–recall curve, execution-time comparison, performance comparison bar chart.

End of Report